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Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Course Context: Transformation techniques for single and multiple random variables

Assumption (as per slides): PDF/CDF of original random variable(s) is known *a priori*

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1. Transformation of a Single Random Variable

We begin with the fundamental problem:

> Given a random variable X with known distribution, and a new random variable defined as

Y = g(X),

]

find the distribution (CDF and PDF) of Y.

1.1 CDF-Based Method (As Derived in Slides)

We start from first principles.

Step 1: Write the CDF of Y

$$F_Y(y) = P(Y \leq y)$$

Since $Y = g(X)$,

$$F_Y(y) = P(g(X) \leq y)$$

At this stage, the derivation depends on the monotonicity of $g(\cdot)$.

Case 1: $g(\cdot)$ is Monotonically Increasing

If g is strictly increasing, then

$$g(X) \leq y \quad \Longleftrightarrow \quad X \leq g^{-1}(y)$$

Therefore,

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This is the key CDF transformation formula.

Step 2: Differentiate to Obtain the PDF

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

(The absolute value accounts for decreasing transformations.)

Case 2: $g(\cdot)$ is Monotonically Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Since derivative is negative, we again obtain:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

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2. Example from Slides

Example: $X \sim \text{Uniform}(-1, 1)$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Identify Transformation

$$y = \sin\left(\frac{\pi}{2} x\right)$$

Since sine is increasing over $x \in (-1, 1)$, we can invert.

Step 2: Invert

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Step 4: Substitute into Formula

Since

$$f_X(x) = \frac{1}{2}, \quad -1 < x < 1$$

we obtain

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Thus,

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1 - y^2}}, \quad -1 < y < 1}$$

This matches the derivation shown in the lecture.

---

3. Function of Two Random Variables

We now move to joint transformations.

Given:

$$Z = X + Y$$

Find the PDF of Z.

3.1 CDF-Based Derivation (As Presented)

Start with:

$$F_Z(z) = P(Z \leq z)$$

Since $Z = X + Y$,

$$F_Z(z) = P(X + Y \leq z)$$

This defines a region in the x, y-plane:

$$\{(x, y) : x + y \leq z\}$$

Step 1: Write as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

From slides, region is converted as:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

Step 2: Differentiate w.r.t. z

To obtain the PDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

We use **Leibniz Rule** (explicitly derived in slides):

If

$$G(z) = \int_{a(z)}^{b(z)} h(x, z)\, dx$$

then

$$\frac{d}{dz} G(z) = h(b(z), z) b'(z) - h(a(z), z) a'(z) + \int_{a(z)}^{b(z)} \frac{\partial}{\partial z} h(x, z)\, dx$$

In our case:

- Lower limit does not depend on z

- Upper limit is $z - x$

Thus derivative gives:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx$$

3.2 Convolution Result (Independence Case)

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Therefore:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the **convolution integral**, exactly as shown in slides.

---

4. Example: Sum of Two Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Using convolution:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-x)^2/(2\sigma^2)} dx$$

Combining exponents:

$$e^{-\frac{x^2 + (z-x)^2}{2\sigma^2}}$$

Completing the square (as in lecture derivation), integral simplifies to Gaussian form.

Final result:

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Thus variance adds under independence.

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5. Example: Sum of Exponential Random Variables

Given:

$$X, Y \sim \text{Exponential}(\lambda)$$

Independent.

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Using convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$
$$= \lambda^2 e^{-\lambda z} \int_0^z dx = \lambda^2 z e^{-\lambda z}$$

Thus:

$$\boxed{f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z > 0}$$

---

6. Example: Difference of Two RVs

For:

$$Z = X - Y$$

CDF:

$$F_Z(z) = P(X - Y \leq z)$$

Region becomes:

$$x - y \leq z$$

Follow identical region-based integration approach as shown in lecture.

---

7. Ratio Example (Mentioned in Slides)

Example shown:

$$Z = \frac{X}{Y}$$

Also example:

$$R = \sqrt{X^2 + Y^2}$$

These require joint transformation methods using Jacobians (structure introduced in lecture outline).

Final Exam-Ready Summary

1️⃣ Single RV Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

2️⃣ Sum of Two RVs

$$F_Z(z) = P(X + Y \leq z)$$

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

3️⃣ Key Results from Lecture

- Uniform + sine transformation → arcsine-type PDF
- Sum of normals → Normal, variance adds
- Sum of exponentials → Gamma-type PDF
- Leibniz rule essential for differentiating integrals

This completes a faithful, structured reconstruction of Lectures 11–12 strictly following the provided slides and derivations.

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CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Source: Lecture Slides (L11_S2_A and L12_S1_A)

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0. Assumptions and Scope (As Stated in Slides)

- The **PDF/CDF of the original random variable(s)** is known *a priori*.
- Objective:
  1. Transformation of a single random variable.
  2. Function of two random variables.
  3. Detailed derivation for the case $Z = X + Y$.

All derivations below strictly follow the structure and reasoning shown in the lecture slides.

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1. Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

where the distribution of X is known.

We aim to derive the **CDF and PDF of Y**.

1.1 CDF-Based Approach (Step-by-Step Derivation)

We begin from the definition:

$$F_Y(y) = P(Y \leq y)$$

Since $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

At this stage, the derivation depends on the monotonicity of $g(\cdot)$, as explicitly handled in the slides.

Case 1: $g(\cdot)$ Monotonically Increasing

If g is strictly increasing:

$$g(X) \leq y \quad \Longleftrightarrow \quad X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \leq g^{-1}(y))$$

Since $F_X(\cdot)$ is known:

$$\boxed{F_Y(y) = F_X(g^{-1}(y))}$$

Deriving the PDF

Differentiate w.r.t. y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ Monotonically Decreasing

If g is strictly decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Since derivative is negative, the final expression again becomes:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

---

2. Example: Uniform Random Variable

Given in slides:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Identify Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

Given:

$$y = \sin\left(\frac{\pi}{2}x\right)$$

Since sine is increasing on this interval, we invert:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Step 4: Apply Transformation Formula

$$f_Y(y) = f_X(x(y)) \left|\frac{dx}{dy}\right|$$

Substituting:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Thus:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

This matches the derivation structure shown in the lecture.

---

3. Function of Two Random Variables

We now move to joint transformations.

Primary illustrative case in lecture:

$$Z = X + Y$$

---

4. Derivation of Distribution of $Z = X + Y$

Step 1: Start from Definition of CDF

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

This corresponds to the region:

$$x + y \leq z$$

in the (x, y)-plane (as shown graphically in slides).

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Region rewritten:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

---

5. Differentiating Using Leibniz Rule

Slides explicitly introduce the Leibniz Rule:

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y)\, dy$$

then

$$\frac{d}{dx} G(x) = h(x, b(x))b'(x) - h(x, a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)\, dy$$

Applying this to:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

Only upper limit depends on z.

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

---

6. Independent Case (Convolution Integral)

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the **convolution formula**, derived step-by-step in lecture.

---

7. Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Substitute Gaussian PDFs into convolution:

$$f_Z(z) = \int \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-x)^2/(2\sigma^2)} dx$$

Combining exponents and simplifying (as shown in slides):

Result:

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Variance adds under independence.

---

8. Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Apply convolution:

$$\begin{aligned} f_Z(z) &= \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx \\ &= \lambda^2 e^{-\lambda z} \int_0^z dx \\ &= \boxed{\lambda^2 z e^{-\lambda z}} \end{aligned}$$

for $z > 0$.

---

9. Other Transformations Mentioned in Slides

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

These are introduced under joint transformation techniques.

Final Exam-Ready Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two RVs

$$f_Z(z) = \int f_{X,Y}(x, z - x)dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)dx$$

Key Lecture Takeaways

- Always begin from CDF definition.
- Carefully determine region in (x, y)-plane.
- Apply Leibniz rule correctly.
- Convolution arises naturally for sums of independent RVs.
- Variances add for independent normals.
- Sum of exponentials produces Gamma-type form.

This completes a structured, faithful, exam-oriented lecture scribe strictly based on the provided lecture slides.

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CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Source: Lecture Slides (L11_S2_A and L12_S1_A)

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1. Outline of the Lecture (As Given in Slides)

1. **Transformation of Random Variables**
   Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for the case:

$$Z = X + Y$$

---

2. Assumption (Explicitly Stated)

- The **PDF or CDF of the original random variable(s)** is known *a priori*.
- The goal is to derive the distribution of a new random variable defined as a function of known ones.

---

3. Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

where the distribution of X is known.

We derive the distribution of Y using the **CDF approach**, exactly as done in the lecture.

3.1 Starting from the Definition of CDF

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

At this stage, the transformation depends on whether $g(\cdot)$ is increasing or decreasing.

Case 1: $g(\cdot)$ Monotonically Increasing

If g is strictly increasing:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This is the CDF of Y.

Deriving the PDF

Differentiate:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Apply chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Hence:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ Monotonically Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y))\frac{d}{dy}g^{-1}(y)$$

Taking absolute value yields the same final formula:

$$\boxed{f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|}$$

---

4. Example: Uniform Distribution Transformation

Given in lecture:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2}X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2}x\right)$$

Since the sine function is increasing on this interval:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Step 4: Apply Transformation Formula

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Thus:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

---

5. Function of Two Random Variables

We now study transformations involving two random variables.

Primary example in lecture:

$$Z = X + Y$$

---

6. Derivation of Distribution of $Z = X + Y$

Step 1: Begin from CDF Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

This represents the region:

$$x + y \leq z$$

in the (x, y)-plane (as illustrated in slides).

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y\le z} f_{X,Y}(x,y)\,dx\,dy$$

Rewrite region:

$$F_Z(z) = \int_{-\infty}^{\infty}\int_{-\infty}^{z-x} f_{X,Y}(x,y)\,dy\,dx$$

---

7. Differentiating Using Leibniz Rule

Slides introduce **Leibniz Rule**:

If

$$G(x) = \int_{a(x)}^{b(x)} h(x,y)\,dy$$

then:

$$\frac{d}{dx}G(x) = h(x,b(x))b'(x) - h(x,a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x}h(x,y)\,dy$$

Applying to:

$$F_Z(z) = \int_{-\infty}^{\infty}\int_{-\infty}^{z-x} f_{X,Y}(x,y)\,dy\,dx$$

Only upper limit depends on z.

Therefore:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x,z-x)\,dx}$$

---

8. Independent Case — Convolution

If X and Y are independent:

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx}$$

This is the **convolution integral** derived in lecture.

---

9. Example: Sum of Two Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Substitute into convolution:

$$f_Z(z) = \int \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-x)^2/(2\sigma^2)} dx$$

After algebraic simplification (as shown in slides):

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Variance adds under independence.

---

10. Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Factor out common term:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

11. Additional Transformations Mentioned

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

Introduced under joint transformation techniques.

Final Exam Summary

Single RV Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two RVs

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Core Logical Flow to Remember

1. Start from **CDF definition**.
2. Identify the correct **region in the plane**.
3. Convert to **double integral**.
4. Apply **Leibniz rule** to differentiate.
5. Simplify using independence if applicable.

This completes the structured lecture scribe faithfully following the provided slides and derivations, suitable for exam revision.

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1. Lecture Outline (As Given in Slides)

1. **Transformation of Random Variables**
   Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for the case:

$$Z = X + Y$$

---

2. Assumption (Explicitly Stated in Slides)

- The **PDF/CDF of the original random variable(s)** is known *a priori*.
- We aim to determine the distribution of a new random variable defined as a function of known random variables.

---

3. Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

where the distribution of X is known.

The derivation in the lecture proceeds strictly from the definition of the CDF.

3.1 CDF-Based Derivation

We start from the definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

At this point, the relationship between $g(X)$ and X depends on whether g is increasing or decreasing.

Case 1: $g(\cdot)$ Monotonically Increasing

If g is strictly increasing:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

Deriving the PDF

Differentiate with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Hence:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ Monotonically Decreasing

If g is strictly decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value gives the same general formula:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

---

4. Example from Lecture: Uniform Distribution

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2}x\right)$$

Since sine is increasing over $(-1, 1)$:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Step 4: Apply Transformation Formula

$$f_Y(y) = f_X(x(y)) \left|\frac{dx}{dy}\right|$$

Substituting:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Thus:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

---

5. Function of Two Random Variables

We now consider transformations involving two random variables.

Primary illustrative example in lecture:

$$Z = X + Y$$

---

6. Derivation of Distribution of $Z = X + Y$

Step 1: Start from the CDF Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

This defines the region:

$$x + y \leq z$$

in the (x, y)-plane.

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite region:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

---

7. Differentiation Using Leibniz Rule

Slides introduce the **Leibniz Rule**:

If

$$G(z) = \int_{a(z)}^{b(z)} h(z, y)\, dy$$

then:

$$\frac{d}{dz} G(z) = h(z, b(z))b'(z) - h(z, a(z))a'(z) + \int_{a(z)}^{b(z)} \frac{\partial}{\partial z} h(z, y)\, dy$$

Applying this to $F_Z(z)$:

- Lower limit does not depend on z.
- Upper limit is $z - x$.

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

---

8. Independent Case — Convolution

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the **convolution integral**.

---

9. Example: Sum of Two Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Substitute Gaussian PDFs into convolution integral:

$$f_Z(z) = \int \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-x)^2/(2\sigma^2)} dx$$

After algebraic simplification (as shown in lecture):

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Thus variance adds.

---

10. Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Factor:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

---

11. Additional Transformations Mentioned

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

These are introduced under joint transformation techniques.

Final Exam Summary

Single RV Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two RVs

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z-x)\, dx$$

Logical Flow to Remember

1. Start from CDF definition.
2. Identify region in (x, y)-plane.
3. Express as double integral.
4. Apply Leibniz rule.
5. Use independence when applicable.

This completes the lecture scribe strictly following the structure, derivations, and examples provided in the lecture slides, formatted for exam revision.

> Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Source: L11_S2_A and L12_S1_A Lecture Slides

---

1. Lecture Structure (As Given in Slides)

The lecture is organized into three main components:

1. **Transformation of Random Variables**
   Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for the case:

$$Z = X + Y$$

---

2. Assumptions (Explicitly Stated)

- The **PDF or CDF of the original random variable(s)** is known *a priori*.
- The objective is to derive the distribution of a new random variable defined as a function of known random variables.

PART I: Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

where the distribution of X is known.

The derivation in the lecture proceeds from the definition of the CDF.

---

3. Derivation Using the CDF Method

We begin with the definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

At this point, the next step depends on whether $g(\cdot)$ is increasing or decreasing.

Case 1: $g(\cdot)$ is Monotonically Increasing

If g is strictly increasing:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This gives the CDF of Y.

Deriving the PDF

Differentiate with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ is Monotonically Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value gives:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Hence, the final transformation formula is valid for both increasing and decreasing monotonic functions.

Example from Slides

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2} x\right)$$

Since sine is increasing over $(-1, 1)$, invert:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Step 4: Apply the Formula

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Therefore:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

PART II: Function of Two Random Variables

We now consider joint transformations.

The primary illustrative example:

$$Z = X + Y$$

---

4. Derivation of Distribution of $Z = X + Y$

Step 1: Start from CDF Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

This defines the region:

$$x + y \leq z$$

in the (x, y)-plane.

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite the region:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

---

5. Differentiation Using Leibniz Rule

Slides introduce the **Leibniz Rule**:

If

$$G(z) = \int_{a(z)}^{b(z)} h(z, y)\, dy$$

then

$$\frac{d}{dz} G(z) = h(z, b(z))b'(z) - h(z, a(z))a'(z) + \int_{a(z)}^{b(z)} \frac{\partial}{\partial z} h(z, y)\, dy$$

Applying to $F_Z(z)$:

- Lower limit does not depend on z.
- Upper limit is $z - x$.

Therefore:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

Independent Case — Convolution

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the convolution integral.

Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Substitute Gaussian PDFs into convolution and simplify (as shown in lecture):

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Variance adds under independence.

Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Factor:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Additional Transformations Mentioned in Slides

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

These are introduced under joint transformation techniques.

Final Exam-Oriented Summary

Single RV Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two RVs

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Logical Flow for Exam

1. Start from the CDF definition.
2. Translate inequality into region.
3. Express as integral.
4. Apply Leibniz rule.
5. Use independence if applicable.

This completes the lecture scribe strictly aligned with the provided slides, formatted for exam revision.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic

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CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Source: L11_S2_A and L12_S1_A Slides

---

1. Lecture Outline (As Presented)

The lecture is structured into three main components:

1. **Transformation of Random Variables**
   Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for:

$$Z = X + Y$$

---

2. Fundamental Assumption

As stated in the slides:

- The **PDF or CDF of the original random variable(s)** is known *a priori*.
- The objective is to derive the distribution of a new random variable defined as a function of known random variables.

PART I — Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

where the distribution of X is known.

The lecture derives the distribution of Y using the **CDF method**.

---

3. CDF-Based Derivation

We begin from the definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

At this point, the inequality must be rewritten in terms of X.
The approach depends on the monotonicity of $g(\cdot)$.

Case 1: $g(\cdot)$ Monotonically Increasing

If g is strictly increasing:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This gives the CDF of Y.

Derivation of the PDF

Differentiate with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Applying the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ Monotonically Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Hence the same transformation formula applies.

Example from Lecture

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2}x\right)$$

Since sine is increasing on $(-1, 1)$:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Step 4: Apply Transformation Formula

$$f_Y(y) = f_X(x(y)) \left|\frac{dx}{dy}\right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Therefore:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

PART II — Function of Two Random Variables

We now consider joint transformations.

Primary example:

$$Z = X + Y$$

---

4. Distribution of $Z = X + Y$

Step 1: Start from CDF Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

This inequality defines the region:

$$x + y \leq z$$

in the (x, y)-plane.

Step 2: Convert to Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite the region:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

---

5. Application of Leibniz Rule

Slides present the **Leibniz Rule**:

If

$$G(z) = \int_{a(z)}^{b(z)} h(z, y)\, dy$$

then

$$\frac{d}{dz} G(z) = h(z, b(z))b'(z) - h(z, a(z))a'(z) + \int_{a(z)}^{b(z)} \frac{\partial}{\partial z} h(z, y)\, dy$$

Applying to $F_Z(z)$:

- Lower limit independent of z.
- Upper limit is $z - x$.

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

Independent Case — Convolution

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the convolution formula.

Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Substituting into convolution and simplifying (as shown in lecture):

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Variance adds under independence.

Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Factor:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Additional Transformations Mentioned

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

Introduced under joint transformation techniques.

Final Exam Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Variables

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Logical Flow to Follow in Exam

1. Begin from CDF definition.

2. Translate inequality to integration region.
3. Express as integral.
4. Apply Leibniz rule.
5. Use independence if applicable.

This completes the lecture scribe strictly aligned with the provided lecture slides and structured for exam preparation.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Source: Lecture Slides (L11_S2_A and L12_S1_A)

---

1. Lecture Outline (As Given in Slides)

The lecture covers:

1. **Transformation of Random Variables**
   Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.

3. **Illustrative Example**
   Detailed derivation for the case:

$$Z = X + Y$$

---

2. Assumption (Explicitly Stated)

- The **PDF or CDF of the original random variable(s)** is known *a priori*.
- Objective: derive the distribution of a new random variable defined as a function of known random variables.

PART I — Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

where the distribution of X is known.

The lecture derives the distribution of Y using the **CDF approach**.

---

3. Derivation Using CDF Definition

Start from the definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

The inequality must now be rewritten in terms of X. This depends on the monotonicity of $g(\cdot)$.

Case 1: $g(\cdot)$ is Monotonically Increasing

If g is strictly increasing:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This gives the CDF of Y.

Deriving the PDF

Differentiate with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Apply the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Hence:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ is Monotonically Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Thus, the final transformation formula holds in both cases.

Example from Lecture

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2}X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2}x\right)$$

Since sine is increasing over $(-1, 1)$:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Step 4: Apply the Formula

$$f_Y(y) = f_X(x(y)) \left|\frac{dx}{dy}\right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Thus:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

PART II — Function of Two Random Variables

We now consider transformations involving two random variables.

Primary example:

$$Z = X + Y$$

---

4. Distribution of $Z = X + Y$

Step 1: Start from CDF Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

This inequality defines the region:

$$x + y \leq z$$

in the (x, y)-plane.

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite region:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

5. Application of Leibniz Rule

Slides introduce the **Leibniz Rule**:

If

$$G(z) = \int_{a(z)}^{b(z)} h(z, y)\, dy$$

then

$$\frac{d}{dz} G(z) = h(z, b(z))b'(z) - h(z, a(z))a'(z) + \int_{a(z)}^{b(z)} \frac{\partial}{\partial z} h(z, y)\, dy$$

Applying to $F_Z(z)$:

- Lower limit does not depend on z.
- Upper limit is $z - x$.

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

Independent Case — Convolution

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the convolution formula.

Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Substituting into convolution and simplifying (as shown in lecture):

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Variance adds under independence.

Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Factor:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Additional Transformations Mentioned

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

These are introduced under joint transformation techniques.

Final Exam-Oriented Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Variables

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Logical Flow for Exam

1. Begin from CDF definition.
2. Translate inequality to integration region.
3. Express as integral.
4. Apply Leibniz rule.
5. Use independence when applicable.

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CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Source: L11_S2_A and L12_S1_A Lecture Slides

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1. Lecture Structure (As Given in Slides)

The lecture is organized into:

1. **Transformation of Random Variables**
   Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for the case:

$$Z = X + Y$$

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2. Fundamental Assumption

As explicitly mentioned:

- The **PDF or CDF of the original random variable(s)** is known *a priori*.
- The goal is to determine the distribution of a new random variable defined as a function of known variables.

PART I — Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

where the distribution of X is known.

The lecture proceeds using the **CDF method**.

3. Derivation Using CDF Definition

Start from:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

The inequality must be expressed in terms of X.
This depends on whether $g(\cdot)$ is increasing or decreasing.

Case 1: $g(\cdot)$ Monotonically Increasing

If g is strictly increasing:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

Deriving the PDF

Differentiate:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ Monotonically Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Example from Slides

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2} x\right)$$

Since sine is increasing over $(-1, 1)$:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Step 4: Apply the Formula

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Therefore:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

PART II — Function of Two Random Variables

We now consider joint transformations.

Primary example:

$$Z = X + Y$$

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4. Distribution of $Z = X + Y$

Step 1: Start from CDF Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

This inequality defines the region:

$$x + y \leq z$$

in the (x, y)-plane.

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite region:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

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5. Application of Leibniz Rule

Slides introduce the **Leibniz Rule**:

If

$$G(z) = \int_{a(z)}^{b(z)} h(z, y)\, dy$$

then

$$\frac{d}{dz} G(z) = h(z, b(z))b'(z) - h(z, a(z))a'(z) + \int_{a(z)}^{b(z)} \frac{\partial}{\partial z} h(z, y)\, dy$$

Applying to $F_Z(z)$:

- Lower limit independent of z.
- Upper limit is $z - x$.

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

Independent Case — Convolution

If X and Y are independent:

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx}$$

This is the convolution formula.

Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Substituting into convolution and simplifying (as shown in lecture):

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Variance adds under independence.

Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Factor:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Additional Transformations Mentioned

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

These are introduced under joint transformation techniques.

Final Exam-Oriented Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Variables

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Logical Flow to Follow in Exam

1. Start from CDF definition.
2. Translate inequality into region.
3. Express as integral.
4. Apply Leibniz rule.
5. Use independence when applicable.

This completes the lecture scribe strictly aligned with the provided lecture slides and structured for exam preparation.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Source: Lecture Slides (L11_S2_A and L12_S1_A)

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1. Lecture Outline (As Presented in Slides)

The lecture consists of:

1. **Transformation of Random Variables**
   Learning transformation techniques for random variables.
2. **Function of Two Random Variables**
   Joint transformations and derived distributions.
3. **Illustrative Example**
   Detailed derivation for the case:

$$Z = X + Y$$

---

2. Assumption

As explicitly stated:

- The **PDF/CDF of the original random variable(s)** is known *a priori*.
- The objective is to derive the distribution of a new random variable defined as a function of known random variables.

PART I — Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

where the distribution of X is known.

The lecture derives the distribution of Y using the **CDF-based method**.

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3. CDF-Based Derivation

Start from the definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

To proceed, we must rewrite the inequality in terms of X.
This depends on whether $g(\cdot)$ is increasing or decreasing.

Case 1: $g(\cdot)$ is Monotonically Increasing

If g is strictly increasing:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This gives the CDF of Y.

Deriving the PDF

Differentiate:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ is Monotonically Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Thus, the same formula applies in both cases.

Example from Slides

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2} x\right)$$

Since sine is increasing on $(-1, 1)$:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 3: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Step 4: Apply the Formula

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Thus:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

PART II — Function of Two Random Variables

We now study joint transformations.

Primary example in lecture:

$$Z = X + Y$$

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4. Distribution of $Z = X + Y$

Step 1: Start from CDF Definition

$$F_Z(z) = P(Z \le z) = P(X + Y \le z)$$

This inequality defines the region:

$$x + y \le z$$

in the (x, y)-plane.

Step 2: Convert to Double Integral

$$F_Z(z) = \iint_{x+y\le z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite region:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

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5. Application of Leibniz Rule

Slides introduce the **Leibniz Rule**:

If

$$G(z) = \int_{a(z)}^{b(z)} h(z, y)\, dy$$

then

$$\frac{d}{dz} G(z) = h(z, b(z))b'(z) - h(z, a(z))a'(z) + \int_{a(z)}^{b(z)} \frac{\partial}{\partial z} h(z, y)\, dy$$

Applying this to $F_Z(z)$:

- Lower limit does not depend on z.
- Upper limit is $z - x$.

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

Independent Case — Convolution

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the convolution formula.

Example: Sum of Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Substitute into convolution and simplify (as shown in lecture):

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Variance adds under independence.

Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

Factor:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Additional Transformations Mentioned

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

Introduced under joint transformation techniques.

Final Exam-Oriented Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Variables

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Logical Flow to Follow in Exam

1. Begin from CDF definition.
2. Translate inequality to integration region.
3. Express as integral.
4. Apply Leibniz rule.
5. Use independence when applicable.

This completes the lecture scribe strictly aligned with the provided lecture slides and formatted for exam preparation.

> Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Source: Lecture Slides L11_S2_A and L12_S1_A

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1. Lecture Objectives (As Given in Slides)

The lecture covers:

1. **Transformation of Random Variables**
   Deriving the distribution of a new random variable defined as a function of a known random variable.
2. **Function of Two Random Variables**
   Deriving the distribution of a function of two jointly distributed random variables.

The derivations proceed rigorously from first principles, primarily using the **CDF method**.

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2. Fundamental Assumption

Throughout the lecture:

- The **PDF/CDF of the original random variable(s)** is known.
- The goal is to determine the distribution of a transformed variable defined as a function of these known variables.

No additional assumptions are introduced beyond those explicitly stated in the slides.

PART I — Transformation of a Single Random Variable

Let

$$Y = g(X)$$

where the distribution of X is known.

The lecture derives the distribution of Y using the **CDF-based approach**, proceeding systematically.

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3. CDF Method for Single Variable Transformation

Step 1: Start from Definition

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation:

$$F_Y(y) = P(g(X) \leq y)$$

The next step depends on whether $g(\cdot)$ is monotonic.

Case 1: $g(\cdot)$ is Strictly Increasing

If g is increasing, then the inequality preserves direction:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This gives the CDF of Y.

Step 2: Obtain PDF by Differentiation

Differentiate with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus,

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

The absolute value ensures correctness for all monotonic cases.

Case 2: $g(\cdot)$ is Strictly Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value yields the same final formula:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Hence, the transformation formula is valid for any strictly monotonic g.

Example from Lecture Slides

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the distribution of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2} x\right)$$

On $(-1, 1)$, the sine function is strictly increasing.

Thus:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 3: Compute Derivative of Inverse

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Step 4: Apply Transformation Formula

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|$$

Substitute values:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Thus:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

This completes the derivation exactly as shown in the slides.

PART II — Function of Two Random Variables

Now consider a function of two random variables.

Primary example from slides:

$$Z = X + Y$$

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4. Distribution of $Z = X + Y$

We derive the distribution directly from first principles.

Step 1: Start from CDF Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

This inequality defines the region:

$$x + y \leq z$$

in the (x, y)-plane.

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite limits explicitly:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

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5. Differentiation Using Leibniz Rule

To obtain the PDF:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using Leibniz rule for differentiation under the integral sign:

- The upper limit depends on z.
- The lower limit does not.

Therefore,

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

This is the general formula.

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6. Independent Case — Convolution

If X and Y are independent:

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

Substitute into the general formula:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx}$$

This is the **convolution integral**.

Example 1: Sum of Two Normal Random Variables

Given:

$$X \sim N(0, \sigma^2), \quad Y \sim N(0, \sigma^2)$$

Independent.

Applying convolution and simplifying as shown in slides:

$$\boxed{Z \sim N(0, 2\sigma^2)}$$

Thus:

- Means add.
- Variances add under independence.

Example 2: Sum of Two Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)}\, dx$$

Factor common terms:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Additional Transformations Mentioned in Slides

The lecture also lists:

- $Z = X - Y$
- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

These are introduced as extensions of joint transformation techniques.

Final Exam-Oriented Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Random Variables

$$f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$$

If independent:

$$f_Z(z) = \int f_X(x) f_Y(z - x)\, dx$$

Logical Flow to Follow in Examination

1. Begin from the CDF definition.
2. Express the event as an inequality.
3. Convert to integral form.
4. Differentiate carefully.
5. Apply independence if given.

This scribe follows the structure, derivations, and scope of Lecture 11–12 exactly as presented in the provided slides, and is intended to serve as precise exam reference material.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

Below is the lecture scribe written strictly from the structure, scope, derivations, and examples contained in Lecture Slides **L11_S2_A** and **L12_S1_A**. The presentation follows the exact logical flow used in the slides and is formatted as exam reference material.

CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

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1. Lecture Outline (As Given in Slides)

The lecture covers:

1. **Transformation of Random Variables**
2. **Function of Two Random Variables**
3. **Distribution of the Sum of Two Random Variables**

The derivations are performed rigorously using the **CDF-based method**, followed by differentiation to obtain the PDF.

PART I — Transformation of a Single Random Variable

2. Problem Setup

Let:

$$Y = g(X)$$

where:

- The distribution (PDF/CDF) of X is known.
- The function $g(\cdot)$ is given.
- The objective is to derive the distribution of Y.

The derivation begins from the definition of the CDF of Y.

---

3. CDF Method

By definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

At this stage, the inequality must be rewritten in terms of X.
The next step depends on whether $g(\cdot)$ is monotonic.

Case 1: $g(\cdot)$ is Strictly Increasing

If g is increasing, then:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This expresses the CDF of Y in terms of the known CDF of X.

Obtaining the PDF

Differentiate with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Apply the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus,

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ is Strictly Decreasing

If g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiate:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value yields the same final expression:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Hence, the same formula applies for any strictly monotonic transformation.

---

4. Example from Slides

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the distribution of:

$$Y = \sin\left(\frac{\pi}{2} X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Check Monotonicity

On the interval $(-1, 1)$,

$$\sin\left(\frac{\pi}{2} x\right)$$

is strictly increasing.

Thus, we use the increasing-case formula.

Step 3: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2} x\right)$$

Taking inverse sine:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 4: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Step 5: Apply Transformation Formula

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Thus,

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

This completes the derivation exactly as shown in the lecture.

PART II — Function of Two Random Variables

Now consider two random variables X and Y with known joint PDF.

We define:

$$Z = X + Y$$

The objective is to find the distribution of Z.

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5. Derivation of Distribution of $Z = X + Y$

Step 1: Start from Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

The inequality defines the region:

$$x + y \leq z$$

in the (x, y)-plane.

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite limits:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

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6. Differentiate to Obtain PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using differentiation under the integral sign (Leibniz rule):

- The lower limit does not depend on z.
- The upper limit is $z - x$.

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

---

7. Independent Case (Convolution)

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the convolution formula.

---

8. Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent and identical.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)}\, dx$$

Factor common terms:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Final Exam Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Variables

General case:

$$f_Z(z) = \int f_{X,Y}(x, z-x)\, dx$$

Independent case:

$$f_Z(z) = \int f_X(x) f_Y(z-x)\, dx$$

Recommended Exam Strategy

1. Begin with the definition of CDF.
2. Convert the probability statement into inequalities.
3. Express as integrals.
4. Differentiate carefully.
5. Apply independence if given.

This scribe strictly follows the derivations, structure, and examples presented in Lecture 11–12 and is formatted for direct use as exam reference material.

> Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

Below is the lecture scribe written strictly according to **Lecture 11–12 (L11_S2_A and L12_S1_A)**. The structure, derivations, logical order, and examples follow the slides faithfully. No additional material beyond the provided lecture context is introduced.

CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

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1. Lecture Outline (As Given in Slides)

The lecture consists of:

1. **Transformation of Random Variables**
2. **Function of Two Random Variables**
3. **Distribution of the Sum of Two Random Variables**

The derivations are performed using the **CDF-based method**, followed by differentiation to obtain the PDF.

PART I — Transformation of a Single Random Variable

2. Problem Statement

Let:

$$Y = g(X)$$

where:

- The distribution of X is known.
- The function $g(\cdot)$ is given.
- The objective is to determine the distribution of Y.

The derivation begins directly from the definition of the CDF.

---

3. CDF-Based Derivation

By definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

At this point, the inequality must be rewritten in terms of X.
This depends on whether $g(\cdot)$ is monotonic.

Case 1: $g(\cdot)$ is Strictly Increasing

If g is strictly increasing, then:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This expresses the CDF of Y entirely in terms of the known CDF of X.

Deriving the PDF

Differentiate with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Apply the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Therefore:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ is Strictly Decreasing

If g is strictly decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiate:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Hence, the same final formula holds for any strictly monotonic transformation.

---

4. Example from Slides

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2}X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Check Monotonicity

On $(-1, 1)$, the function:

$$\sin\left(\frac{\pi}{2}x\right)$$

is strictly increasing.

Thus, the increasing-case formula applies.

Step 3: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2}x\right)$$

Taking inverse sine:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 4: Compute Derivative of the Inverse

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Step 5: Apply Transformation Formula

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Thus:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

This completes the derivation exactly as presented in the lecture.

PART II — Function of Two Random Variables

Now consider two random variables X and Y with joint PDF $f_{X,Y}(x, y)$.

We define:

$$Z = X + Y$$

The goal is to determine the distribution of Z.

---

5. Derivation of Distribution of $Z = X + Y$

Step 1: Start from Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

The inequality:

$$x + y \leq z$$

defines a region in the (x, y)-plane.

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite explicitly:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

---

6. Differentiate to Obtain PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using differentiation under the integral sign (Leibniz rule):

- The lower limit does not depend on z.
- The upper limit is $z - x$.

Thus:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

---

7. Independent Case (Convolution)

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute into the general formula:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the convolution integral.

---

8. Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent and identical.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)}\, dx$$

Factor common terms:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Final Exam Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Random Variables

General case:

$$f_Z(z) = \int f_{X,Y}(x, z-x)\, dx$$

Independent case:

$$f_Z(z) = \int f_X(x) f_Y(z-x)\, dx$$

Logical Flow for Examination

1. Start from CDF definition.
2. Express inequality clearly.
3. Convert to integral.
4. Differentiate carefully.
5. Apply independence if given.

This scribe follows exactly the structure, reasoning, and examples provided in Lecture 11–12 and is formatted for precise exam revision.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

Below is the lecture scribe prepared strictly according to **Lecture 11–12 (L11_S2_A and L12_S1_A)**. The structure, logical development, assumptions, derivations, and examples follow the lecture material faithfully. No external material or additional examples are introduced.

CSE400 — Fundamentals of Probability in Computing

Lecture 11–12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

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1. Lecture Outline (As Presented in Slides)

The lecture covers:

1. **Transformation of Random Variables**
2. **Function of Two Random Variables**
3. **Distribution of the Sum of Two Random Variables**

All derivations are carried out using the **CDF method**, followed by differentiation to obtain the PDF.

PART I — Transformation of a Single Random Variable

2. Problem Setup

Let:

$$Y = g(X)$$

where:

- The distribution of X is known.
- The function $g(\cdot)$ is specified.
- The objective is to derive the distribution of Y.

The derivation begins from first principles.

---

3. CDF-Based Derivation

By definition:

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation:

$$F_Y(y) = P(g(X) \leq y)$$

At this point, the probability must be rewritten in terms of X.
The approach depends on whether $g(\cdot)$ is monotonic.

Case 1: $g(\cdot)$ is Strictly Increasing

If g is strictly increasing, then the inequality direction is preserved:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

This expresses the CDF of Y in terms of the known CDF of X.

Deriving the PDF

Differentiate with respect to y:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Apply the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus,

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Case 2: $g(\cdot)$ is Strictly Decreasing

If g is strictly decreasing, the inequality reverses:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Therefore:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Differentiate:

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

Taking absolute value yields:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|}$$

Hence, the same final transformation formula applies in both cases.

---

4. Example from Lecture Slides

Given:

$$X \sim \text{Uniform}(-1, 1)$$

Find the PDF of:

$$Y = \sin\left(\frac{\pi}{2}X\right)$$

Step 1: Known PDF of X

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 2: Check Monotonicity

On $(-1, 1)$,

$$\sin\left(\frac{\pi}{2}x\right)$$

is strictly increasing.

Thus, we apply the increasing-case formula.

Step 3: Invert the Transformation

$$y = \sin\left(\frac{\pi}{2}x\right)$$

Taking inverse sine:

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 4: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Step 5: Apply Transformation Formula

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Thus:

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1}$$

This matches the result derived in the slides.

PART II — Function of Two Random Variables

Now consider two random variables X and Y with known joint PDF $f_{X,Y}(x, y)$.

Define:

$$Z = X + Y$$

The objective is to derive the distribution of Z.

---

5. Derivation of Distribution of $Z = X + Y$

Step 1: Start from Definition

$$F_Z(z) = P(Z \leq z) = P(X + Y \leq z)$$

The inequality:

$$x + y \leq z$$

defines a region in the (x, y)-plane.

Step 2: Express as Double Integral

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y)\, dx\, dy$$

Rewrite explicitly:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x, y)\, dy\, dx$$

---

6. Differentiate to Obtain PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using differentiation under the integral sign:

- Lower limit does not depend on z.
- Upper limit is $z - x$.

Therefore:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx}$$

---

7. Independent Case (Convolution)

If X and Y are independent:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Substitute:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

This is the convolution formula.

---

8. Example: Sum of Exponential Random Variables

Given:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

Independent and identical.

Apply convolution:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)}\, dx$$

Factor:

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$= \boxed{\lambda^2 z e^{-\lambda z}, \quad z > 0}$$

Final Exam Reference Summary

Single Variable Transformation

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Sum of Two Random Variables

General case:

$$f_Z(z) = \int f_{X,Y}(x, z-x)\, dx$$

Independent case:

$$f_Z(z) = \int f_X(x) f_Y(z-x)\, dx$$

Logical Flow to Follow in Examination

1. Begin from the CDF definition.
2. Convert the event into inequalities.
3. Express as integrals.
4. Differentiate carefully.
5. Apply independence when applicable.

This document follows exactly the structure, derivations, and examples contained in Lecture 11–12 and is suitable for exam preparation and revision.